

Polymer Plant Scheduling — Dedicated PE & PP Production Trains

Manufacturing scheduling case study · Julia + JuMP + HiGHS · companion to `notebooks/polymer_pe_pp_scheduling.jl`

This report documents a **production scheduling model** for a polymer plant that manufactures several grades of **polyethylene (PE)** and **polypropylene (PP)**. PE and PP are made on **physically separate trains**: different reactor technology and catalyst systems (e.g. gas-phase/slurry PE reactors vs. Ziegler-Natta/metallocene PP reactors) mean a PE line can never run a PP grade or vice versa. So this is really **two independent single-line scheduling problems, running in parallel** — one per train.

Within a train, switching between grades still costs time and money: color/additive changes and partial purges. The scheduler below decides the order in which each train runs its confirmed orders so as to minimize total changeover cost plus tardiness cost, per train. Each train's sub-problem is a mixed-integer linear program (MILP), built with [JuMP.jl](#) and solved with the open-source [HiGHS](#) solver.

1. Plant data: grades, families, and rates

Five grades run on the plant's two trains: three PE grades (HDPE, LLDPE, LDPE) on the PE train, and two PP grades (homopolymer and copolymer) on the PP train.

Grade	Train	Rate (t/h)
HDPE-5502	PE	5.0
LLDPE-2020	PE	4.5
LDPE-1922	PE	4.0
PP-Homo-1100	PP	5.5
PP-Copo-3300	PP	4.8

2. Customer orders → production lots

A plant does not run one production campaign per purchase order. Sales orders for the **same grade** shipping close together get pooled into one production lot, sized to cover all of them, timed to the **earliest** ship date in the group — producing early never hurts the later orders in the group, it just means finished goods sit in the warehouse a few extra hours. Fewer, larger lots also mean fewer changeovers.

This run uses a **24-hour** consolidation window (same-grade orders due within that many hours of the earliest one in the group are pooled together).

Customer	Grade	Qty (t)	Due (h)
A1	HDPE-5502	45	28
A2	HDPE-5502	35	30
A3	HDPE-5502	40	33
B1	LLDPE-2020	50	38
B2	LLDPE-2020	30	42
C1	PP-Homo-1100	60	52
C2	PP-Homo-1100	40	57
D1	LDPE-1922	25	62
D2	LDPE-1922	35	68
E1	PP-Copo-3300	50	76
E2	PP-Copo-3300	40	84
A4	HDPE-5502	30	90
A5	HDPE-5502	40	98
C3	PP-Homo-1100	70	105
C4	PP-Homo-1100	40	112

15 customer orders consolidate into **7 production lots**:

Lot	Grade	Qty (t)	Due (h)	Weight	Combines
1	HDPE-5502	120	28	1.0	3 customer order(s)
2	HDPE-5502	70	90	1.0	2 customer order(s)
3	LLDPE-2020	80	38	1.0	2 customer order(s)
4	PP-Homo-1100	100	52	1.5	2 customer order(s)
5	PP-Homo-1100	110	105	1.5	2 customer order(s)
6	LDPE-1922	60	62	1.0	2 customer order(s)
7	PP-Copo-3300	90	76	1.5	2 customer order(s)

3. Sequence-dependent changeovers (within a train)

Because a train only ever runs grades from one family, the only changeover decision is *within* that family. Real purge/transition times depend on the **specific grade pair**, not just "same grade or not" — and they are often **asymmetric**: moving to a lighter color or lower-additive grade purges faster than the reverse, because residual material from the *previous* grade is what has to be flushed out. Hours shown are from (row) → to (column):

PE train

from \ to	HDPE-5502	LLDPE-2020	LDPE-1922
HDPE-5502	0.5	2.0	3.5
LLDPE-2020	2.5	0.5	2.0
LDPE-1922	3.0	2.5	0.5

PP train

from \ to	PP-Homo-1100	PP-Copo-3300
PP-Homo-1100	0.5	3.0
PP-Copo-3300	2.0	0.5

There is no PE ↔ PP transition to model — that changeover never happens, because it would require re-equipping the whole train.

Cost rates used in this run: changeover **\$450/h**, tardiness **\$300/h** per unit order weight. (The companion Pluto notebook exposes both as live sliders.)

4. MILP formulation (applied once per train)

Each train's orders are scheduled independently. Node 0 is a dummy "train idle" start state; nodes 1..m are that train's orders. Binary variables x_{ij} indicate that order j runs immediately after node i. Continuous variables C_i and T_i are the completion time and tardiness of order i.

```
minimize    c_co * sum_{i != j} s_ij * x_ij  +  c_tar * sum_i w_i * T_i

subject to  sum_j x_0j = 1
            sum_{i != j} x_ij = 1                                for all j
```

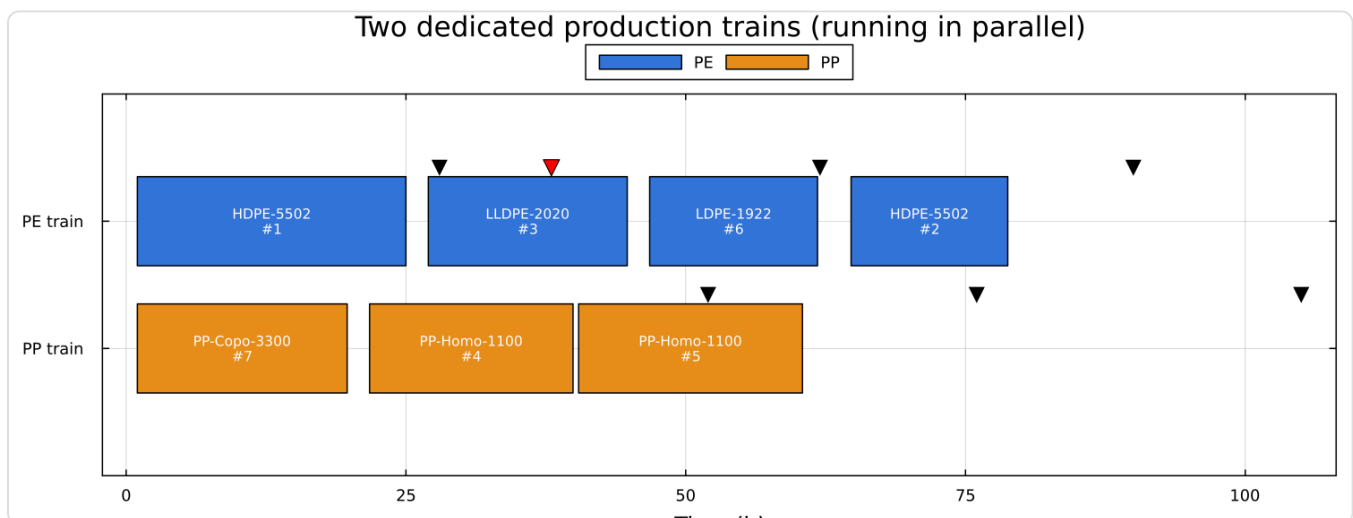
$$\begin{aligned} \sum_{j \neq i} x_{ij} &\leq 1 && \text{for all } i \\ C_j &\geq C_i + s_{ij} + p_j - M(1 - x_{ij}) && \text{for all } i \neq j \\ T_i &\geq C_i - d_i, \quad T_i \geq 0 \\ x_{ij} &\in \{0,1\}, \quad C_i, T_i \geq 0 \end{aligned}$$

Completion times must strictly increase along any chain of active arcs, so this "big-M timing" formulation rules out sub-tours on its own — no separate subtour-elimination constraints (e.g. MTZ) are needed. The two trains' sub-problems share no variables, so solving them independently with HiGHS is exact, not a heuristic decomposition — both reach global optimality in well under a second.

5. Optimal campaign sequence, per train

Train	Order	Grade	Qty (t)	Start (h)	End (h)	Due (h)	Status
PE	1	HDPE-5502	120	1.0	25.0	28.0	on time
PE	3	LLDPE-2020	80	27.0	44.8	38.0	6.8 h late
PE	6	LDPE-1922	60	46.8	61.8	62.0	on time
PE	2	HDPE-5502	70	64.8	78.8	90.0	on time
PP	7	PP-Copo-3300	90	1.0	19.8	76.0	on time
PP	4	PP-Homo-1100	100	21.8	39.9	52.0	on time
PP	5	PP-Homo-1100	110	40.4	60.4	105.0	on time

6. Gantt chart



Grade blocks are colored by polymer family (**PE** vs. **PP**); triangles mark each order's due date (red = missed). Because the two trains never compete for the same equipment, they run **simultaneously**: overall makespan is the *slower* of the two trains, not the sum of both.

7. Key performance indicators

PE TRAIN MAKESPAN	PP TRAIN MAKESPAN	OVERALL MAKESPAN	TOTAL COST
78.8 h	60.4 h	78.8 h	\$7208

Tardiness: 6.8 h across 1 of 7 orders. PE train cost: \$5633; PP train cost: \$1575.

8. Discussion & extensions

Modeling PE and PP as two **dedicated, parallel trains** — rather than one shared line that pays a family-changeover penalty — matters economically: it removes a changeover that can never actually happen on real equipment, and it lets both product families progress at once instead of queueing behind each other. Natural next steps for a production-grade version:

- **More than one train per family** — e.g. two PE lines with different capability sets; extend the model to a proper parallel-machine assignment problem within a family.
- **Minimum/maximum campaign length** — avoid uneconomically short runs by bounding the number of consecutive orders of the same grade.
- **Data-driven changeover matrix** — populate the grade-pair matrix from real historical purge-time records (e.g. a CSV export from the plant historian) instead of hand-entered estimates.
- **Shared upstream/downstream utilities** — if both trains draw from a common feedstock or packaging line, that shared resource needs its own capacity constraint linking the two schedules.
- **Rolling-horizon re-optimization** — re-solve as new orders arrive or priorities change, warm-starting from the current sequence.
- **Robustness** — sample uncertain processing rates / late raw-material arrivals and re-optimize, or solve a chance-constrained variant.

Because each train's model is a compact, independent JuMP program, all of the above are incremental changes to the scheduling function above, not a rewrite. An interactive version of this same model — with live cost-weight sliders — is available as a Pluto.jl notebook at [notebooks/polymer_pe_pp_scheduling.jl](#) in this repository.

